



ALMANAC

Engineering Handbook

Shiv Vyas · Revision E

Almanac is a personal iOS app that gathers every signal from the devices you already wear and carry (Apple Watch, iPhone, Whoop, Fitbit, a Renpho scale, your bank, your calendar) into one local store, scores your life across nine sectors, and puts a coach called LIFO on top of all of it. This handbook is the single reference for how it is built, why it is built that way, what keeps it fast and cheap, and what it does when things go wrong. It is written to be studied: every chapter ends with a reading order through the code and questions to check yourself against.

REVISION E 15 SEPTEMBER 2026 IOS 26 · WATCHOS 26 · SWIFT 6 ARCHITECTURE: MVVM
STATUS: TESTFLIGHT BUILD 17 AUDIENCE: THE AUTHOR

Table of contents

Every line is a link as well as a page number. Part two is the features, one chapter per screen, each in the same shape.

PART ONE: FOUNDATIONS

1	Stub ch01	4
1.1	Intent	4
1.2	Study notes	4
2	Stub ch02	5
2.1	Intent	5
2.2	Study notes	5
3	Stub ch03	6
3.1	Intent	6
3.2	Study notes	6
4	Stub ch04	7
4.1	Intent	7
4.2	Study notes	7

PART TWO: THE SCREENS

5	Stub ch05	9
5.1	Intent	9
5.2	Study notes	9
6	Stub ch06	10
6.1	Intent	10
6.2	Study notes	10
7	Stub ch07	11
7.1	Intent	11
7.2	Study notes	11
8	Stub ch08	12
8.1	Intent	12
8.2	Study notes	12
9	Stub ch09	13
9.1	Intent	13
9.2	Study notes	13
10	Stub ch10	14
10.1	Intent	14
10.2	Study notes	14
11	Stub ch11	15
11.1	Intent	15
11.2	Study notes	15
12	Stub ch12	16
12.1	Intent	16
12.2	Study notes	16
13	Stub ch13	17
13.1	Intent	17
13.2	Study notes	17
14	Stub ch14	18
14.1	Intent	18
14.2	Study notes	18

15	Stub ch15	19
15.1	Intent	19
15.2	Study notes	19
16	Stub ch16	20
16.1	Intent	20
16.2	Study notes	20
17	Stub ch17	21
17.1	Intent	21
17.2	Study notes	21
18	Stub ch18	22
18.1	Intent	22
18.2	Study notes	22

PART THREE: UNDER THE HOOD

19	Stub ch19	24
19.1	Intent	24
19.2	Study notes	24
20	Stub ch20	25
20.1	Intent	25
20.2	Study notes	25
21	Stub ch21	26
21.1	Intent	26
21.2	Study notes	26
22	Stub ch22	27
22.1	Intent	27
22.2	Study notes	27
23	Stub ch23	28
23.1	Intent	28
23.2	Study notes	28
24	Stub ch24	29
24.1	Intent	29
24.2	Study notes	29

APPENDICES: REFERENCE

A	Stub chA	31
A.1	Intent	31
A.2	Study notes	31
B	Stub chB	32
B.1	Intent	32
B.2	Study notes	32
C	Stub chC	33
C.1	Intent	33
C.2	Study notes	33

PART ONE

Foundations

The shape of the system before any screen: how the modules depend on each other, where state lives, how a screen is built, what every screen is built from, and how a person gets through the door.

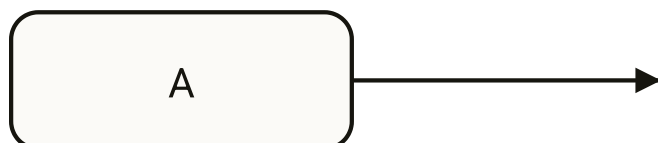
- 1. Stub ch01
- 2. Stub ch02
- 3. Stub ch03
- 4. Stub ch04

1 Stub ch01

Stub lede for ch01 & friends.

1.1 Intent

Stub text with a link and `<code>` .



Stub.

1.2 Study notes

READ IN THIS ORDER

1. one

Q?

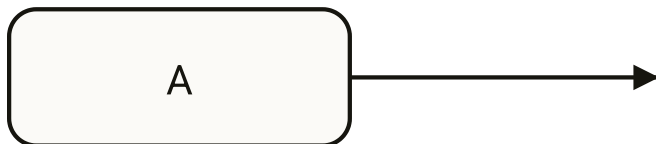
A.

2 Stub ch02

Stub lede for ch02 & friends.

2.1 Intent

Stub text with a link and `<code>` .



Stub.

2.2 Study notes

READ IN THIS ORDER

1. one

Q?

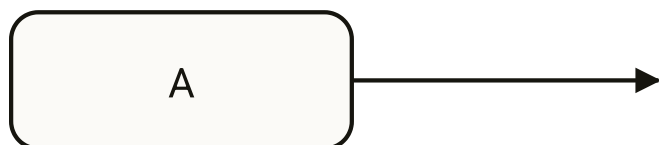
A.

3 Stub ch03

Stub lede for ch03 & friends.

3.1 Intent

Stub text with a link and `<code>` .



Stub.

3.2 Study notes

READ IN THIS ORDER

1. one

Q?

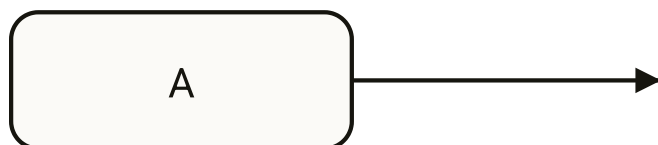
A.

4 Stub ch04

Stub lede for ch04 & friends.

4.1 Intent

Stub text with a link and `<code>` .



Stub.

4.2 Study notes

READ IN THIS ORDER

1. one

Q?

A.

PART TWO

The screens

One chapter per feature. Each follows the same shape: intent, files, model, view model, view, how it works end to end, what keeps it fast and cheap, what breaks and how it is handled, and how to study it.

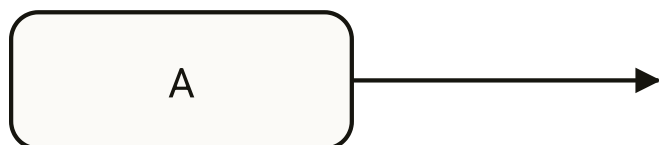
- 5. Stub ch05
- 6. Stub ch06
- 7. Stub ch07
- 8. Stub ch08
- 9. Stub ch09
- 10. Stub ch10
- 11. Stub ch11
- 12. Stub ch12
- 13. Stub ch13
- 14. Stub ch14
- 15. Stub ch15
- 16. Stub ch16
- 17. Stub ch17
- 18. Stub ch18

5 Stub ch05

Stub lede for ch05 & friends.

5.1 Intent

Stub text with a link and `<code>` .



Stub.

5.2 Study notes

READ IN THIS ORDER

1. one

Q?

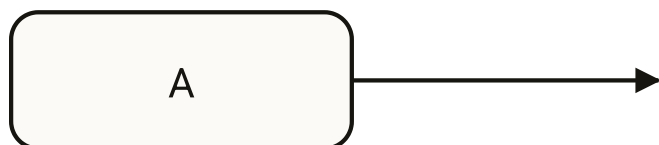
A.

6 Stub ch06

Stub lede for ch06 & friends.

6.1 Intent

Stub text with a link and `<code>` .



Stub.

6.2 Study notes

READ IN THIS ORDER

1. one

Q?

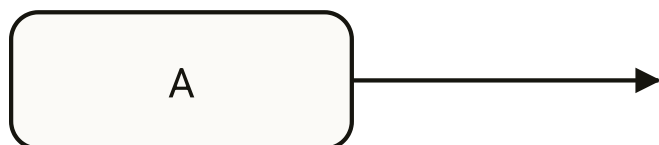
A.

7 Stub ch07

Stub lede for ch07 & friends.

7.1 Intent

Stub text with a link and `<code>` .



Stub.

7.2 Study notes

READ IN THIS ORDER

1. one

Q?

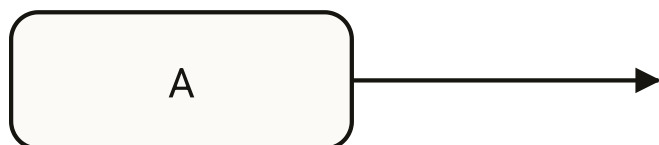
A.

8 Stub ch08

Stub lede for ch08 & friends.

8.1 Intent

Stub text with a link and `<code>` .



Stub.

8.2 Study notes

READ IN THIS ORDER

1. one

Q?

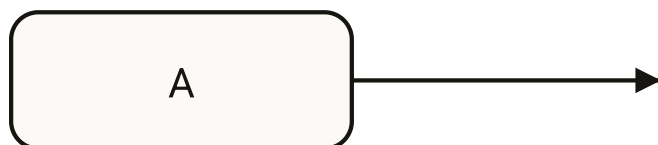
A.

9 Stub ch09

Stub lede for ch09 & friends.

9.1 Intent

Stub text with a link and `<code>` .



Stub.

9.2 Study notes

READ IN THIS ORDER

1. one

Q?

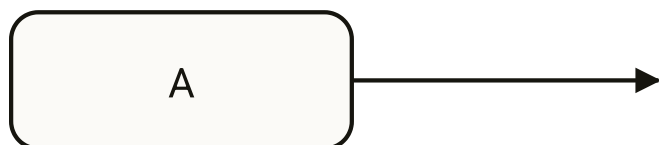
A.

10 Stub ch10

Stub lede for ch10 & friends.

10.1 Intent

Stub text with a link and `<code>` .



Stub.

10.2 Study notes

READ IN THIS ORDER

1. one

Q?

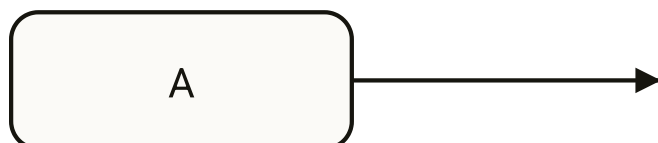
A.

11 Stub ch11

Stub lede for ch11 & friends.

11.1 Intent

Stub text with a link and `<code>` .



Stub.

11.2 Study notes

READ IN THIS ORDER

1. one

Q?

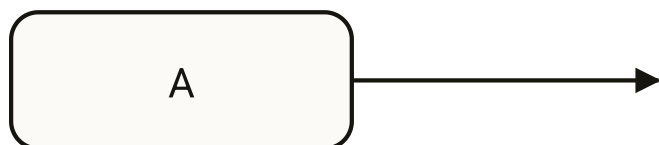
A.

12 Stub ch12

Stub lede for ch12 & friends.

12.1 Intent

Stub text with a link and `<code>` .



Stub.

12.2 Study notes

READ IN THIS ORDER

1. one

Q?

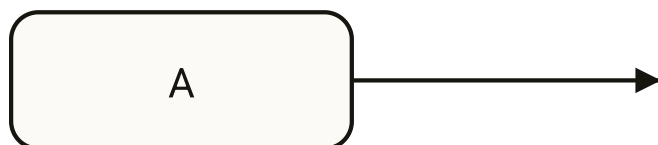
A.

13 Stub ch13

Stub lede for ch13 & friends.

13.1 Intent

Stub text with a link and `<code>` .



Stub.

13.2 Study notes

READ IN THIS ORDER

1. one

Q?

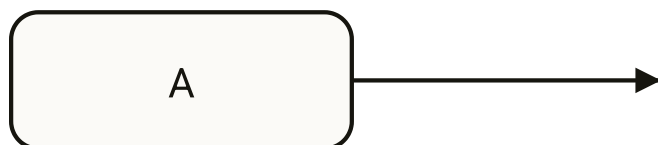
A.

14 Stub ch14

Stub lede for ch14 & friends.

14.1 Intent

Stub text with a link and `<code>` .



Stub.

14.2 Study notes

READ IN THIS ORDER

1. one

Q?

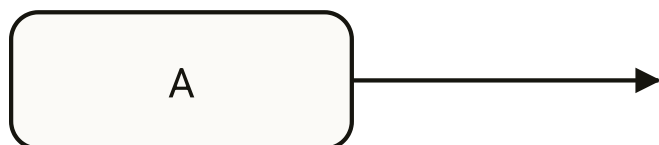
A.

15 Stub ch15

Stub lede for ch15 & friends.

15.1 Intent

Stub text with a link and `<code>` .



Stub.

15.2 Study notes

READ IN THIS ORDER

1. one

Q?

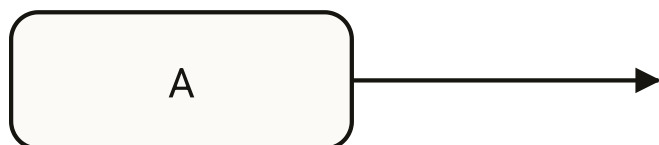
A.

16 Stub ch16

Stub lede for ch16 & friends.

16.1 Intent

Stub text with a link and `<code>` .



Stub.

16.2 Study notes

READ IN THIS ORDER

1. one

Q?

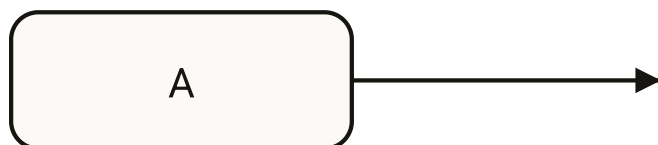
A.

17 Stub ch17

Stub lede for ch17 & friends.

17.1 Intent

Stub text with a link and `<code>` .



Stub.

17.2 Study notes

READ IN THIS ORDER

1. one

Q?

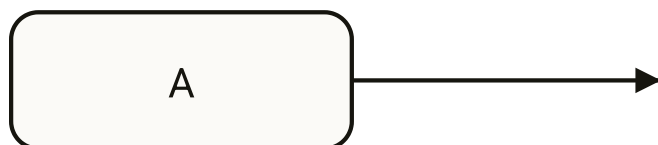
A.

18 Stub ch18

Stub lede for ch18 & friends.

18.1 Intent

Stub text with a link and `<code>` .



Stub.

18.2 Study notes

READ IN THIS ORDER

1. one

Q?

A.

PART THREE

Under the hood

The external systems, the backend, every stored type, the edge cases the app is built to survive, how to set it up and ship it, and what it costs.

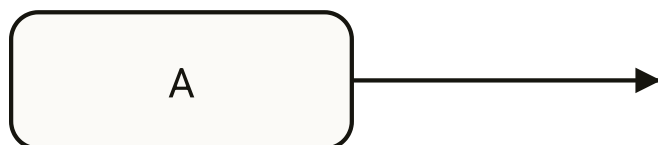
- 19. Stub ch19
- 20. Stub ch20
- 21. Stub ch21
- 22. Stub ch22
- 23. Stub ch23
- 24. Stub ch24

19 Stub ch19

Stub lede for ch19 & friends.

19.1 Intent

Stub text with a link and `<code>` .



Stub.

19.2 Study notes

READ IN THIS ORDER

1. one

Q?

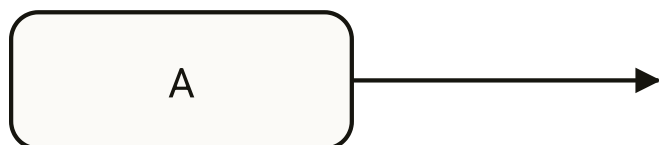
A.

20 Stub ch20

Stub lede for ch20 & friends.

20.1 Intent

Stub text with a link and `<code>` .



Stub.

20.2 Study notes

READ IN THIS ORDER

1. one

Q?

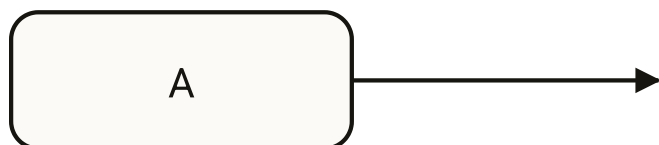
A.

21 Stub ch21

Stub lede for ch21 & friends.

21.1 Intent

Stub text with a link and `<code>` .



Stub.

21.2 Study notes

READ IN THIS ORDER

1. one

Q?

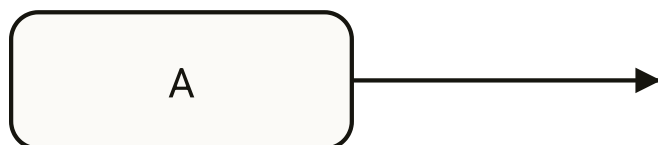
A.

22 Stub ch22

Stub lede for ch22 & friends.

22.1 Intent

Stub text with a link and `<code>` .



Stub.

22.2 Study notes

READ IN THIS ORDER

1. one

Q?

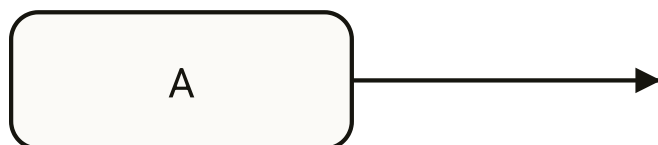
A.

23 Stub ch23

Stub lede for ch23 & friends.

23.1 Intent

Stub text with a link and `<code>` .



Stub.

23.2 Study notes

READ IN THIS ORDER

1. one

Q?

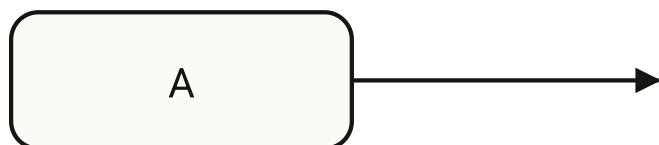
A.

24 Stub ch24

Stub lede for ch24 & friends.

24.1 Intent

Stub text with a link and `<code>` .



Stub.

24.2 Study notes

READ IN THIS ORDER

1. one

Q?

A.

APPENDICES

Reference

Why the product exists, every decision that shaped it, and a glossary with a reading order for studying the code.

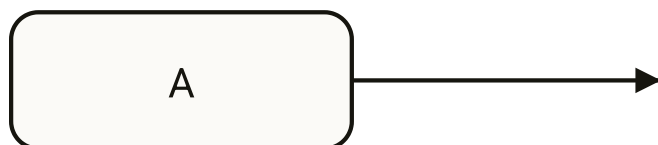
- A. Stub chA
- B. Stub chB
- C. Stub chC

A Stub chA

Stub lede for chA & friends.

A.1 Intent

Stub text with a link and `<code>` .



Stub.

A.2 Study notes

READ IN THIS ORDER

1. one

Q?

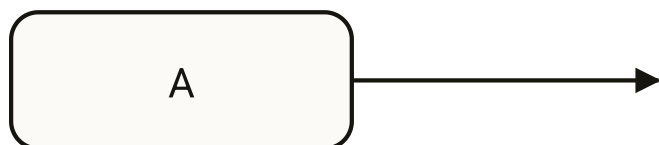
A.

B Stub chB

Stub lede for chB & friends.

B.1 Intent

Stub text with a link and `<code>` .



Stub.

B.2 Study notes

READ IN THIS ORDER

1. one

Q?

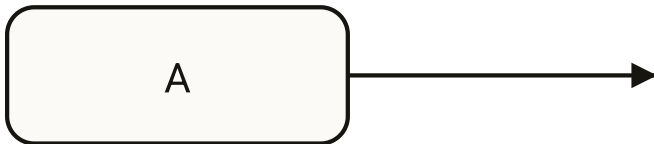
A.

C Stub chC

Stub lede for chC & friends.

C.1 Intent

Stub text with a link and `<code>` .



Stub.

C.2 Study notes

READ IN THIS ORDER

1. one

Q?

A.

Almanac Engineering Handbook, Revision E, 15 September 2026. Written by Shiv Vyas from the code on `main` . Source of record: the code; `docs/superpowers/specs/` for work not yet built. Built by `docs/handbook/build.py` ; page numbers by `render-pdf.sh` ; the Kindle edition by `render-epub.sh` .